

# Point of Care: Modernizing Bedside Charting for Rural Hospitals with HL7v2 Interoperability

## DataBridge

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# Introduction

- Rural hospitals faces challenges with legacy EHR systems, poor interoperability, and inefficient documentation workflows
- Core Idea: A browser based bedside documentation, system, tablet-friendly, that captures patient info and transmits it in a standardized format to an EHR intake service

# Problem Statement

- Healthcare orgs exchange patient info across systems of varying age/capability.
- Rural hospitals: limited resources + growing EHR functionality gap
- Relevant locally -> rural WV hospitals face this directly

# Background/Literature

- 2026 study (15 years of hospital data, 2008-2023): rural hospitals still behind on EHR interoperability
- Gap peaked at 25+ percentage points (2016), still significant
- HL7v2 is the main clinical data exchange standard since 1989 still widely used.

# Proposed System Features

- Tablet-optimized frontend + Python/FastAPI backend
- One page, fast entry, auto save, next-shift nurse gets immediate access to current data
- Admin dashboard (census + alerts), EHR receiver dashboard, audit-log interface

# Timeline

| Week  | Dates         | Task                                                                                                                                                                                                |
|-------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-3   | 08/18 - 09/08 | Brainstorm and develop the project proposal, identify key concepts and unknown aspects, establish objectives and expected outcomes, and refine the concept through collaboration and peer feedback. |
| 4-5   | 09/09 - 09/23 | Develop and refine project requirements, conduct requirements gathering, research existing systems, and begin the Software Requirements Specification (SRS).                                        |
| 6-7   | 09/24 - 10/07 | Continue developing the SRS, define functional and non-functional requirements according to IEEE standards, conduct peer review, and finalize the project's requirements and technical scope.       |
| 8-9   | 10/08 - 10/21 | Complete and submit the SRS; begin initial system architecture, database design, frontend planning, authentication, and RBAC implementation.                                                        |
| 10-11 | 10/22 - 11/04 | Develop the bedside documentation interface, backend API, database functionality, validation, and initial HL7v2 interoperability proof of concept.                                                  |
| 12-13 | 11/05 - 11/18 | Finalize system architecture and HLD documentation, implement and test the simulated EHR receiver, and evaluate professional, ethical, societal, and sustainability considerations.                 |
| 14-15 | 11/19 - 12/02 | Continue system integration, testing, usability improvements, security features, audit logging, and secondary features as time permits.                                                             |
| 16-17 | 12/03 - 12/15 | Final testing, documentation, presentation preparation, system demonstration, project evaluation, and final report preparation.                                                                     |

# Budget and Resources

| Paid Item / Service                   | What we'd use it for                                 | Estimated Cost   |
|---------------------------------------|------------------------------------------------------|------------------|
| <b>Auth0</b>                          | Authentication, MFA, and RBAC                        | <b>\$35</b>      |
| <b>AWS</b>                            | Backend/cloud hosting                                | <b>\$60</b>      |
| <b>Supabase Pro</b>                   | Database, storage, and backups                       | <b>\$25</b>      |
| <b>Sentry</b>                         | Application monitoring and security alerts           | <b>\$26</b>      |
| <b>Cloudflare Pro</b>                 | WAF, DDoS protection, HTTPS/security                 | <b>\$25</b>      |
| <b>1Password Business</b>             | Secure management of development credentials/secrets | <b>\$8</b>       |
| <b>Domain name</b>                    | Professional project website                         | <b>\$15</b>      |
| <b>Security testing service/tools</b> | Additional vulnerability/security testing            | <b>\$250</b>     |
| <b>Cloud backup/storage</b>           | Additional backup space                              | <b>\$20</b>      |
| <b>Contingency</b>                    | Unexpected expenses                                  | <b>\$Vary</b>    |
| <b>TOTAL</b>                          |                                                      | <b>≈ \$450-?</b> |

# References

- [1] A. J. Anzalone, Y. Liu, E. Reisher, E. Frankel, J. R. Hansen, and C. R. Geary, "The digital divide in adoption and advanced use of electronic health records among US hospitals: rural versus urban disparities from 2008 to 2023," *J. Am. Med. Inform. Assoc.*, 2026, Art. no. ocag043. doi: 10.1093/jamia/ocag043.
- [2] Health Level Seven International, "Version 2," HL7 Blog. [Online]. Available: <https://blog.hl7.org/topic/version-2>
- [3] MEDITECH, "EHR Solutions for Rural Health." [Online]. Available: <https://ehr.meditech.com/ehr-solutions/rural-health>

# Git Repo QR code for Scan

- [github.com/AlecHwvu/CapstoneDataBridgeFileMaint.git](https://github.com/AlecHwvu/CapstoneDataBridgeFileMaint.git)

